SOC ANALYST PROJECT SHOWCASE

Splunk SIEM for Windows Security Monitoring, Detection & Investigation

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Built an end-to-end SOC home lab that collected Windows Security telemetry into Splunk, engineered detections and scheduled alerts, visualized security activity in a SOC dashboard, and investigated a triggered authentication alert through incident closure.

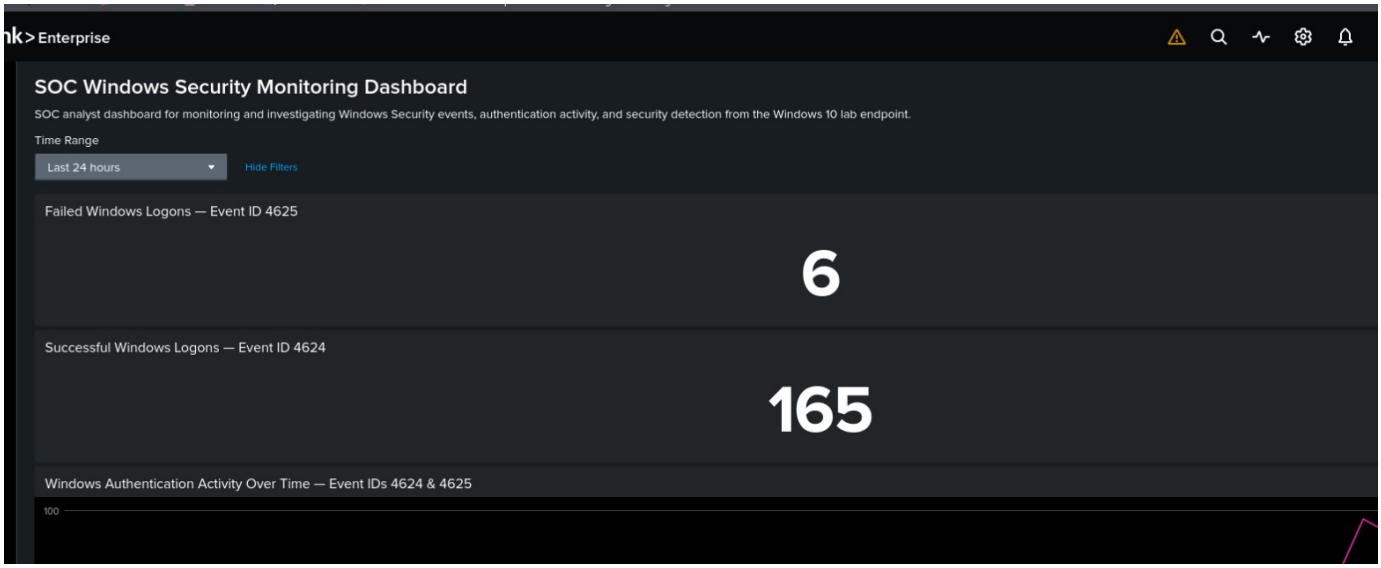
What I Built

- **SIEM Architecture:** Splunk Enterprise 10.4.3 on Ubuntu Server, Windows 10 monitored endpoint, Splunk Universal Forwarder, and Kali Linux analyst workstation in VMware Workstation.
- **Windows Security Monitoring:** Centralized Windows Security Event Logs and validated continuous endpoint-to-SIEM ingestion over TCP 9997.
- **Detection Engineering:** Created a rule for 3 failed Windows logons within 5 minutes using Event ID 4625; refined account-field handling and added per-user alert suppression.
- **SOC Dashboard:** Built monitoring views for authentication, privileged logons, account management, process creation, and PowerShell-related activity.
- **Alert Triage & Investigation:** Validated automatic alert firing, correlated failed Event ID 4625 activity with successful Event ID 4624 authentication, and reviewed Event ID 4688 process creation.

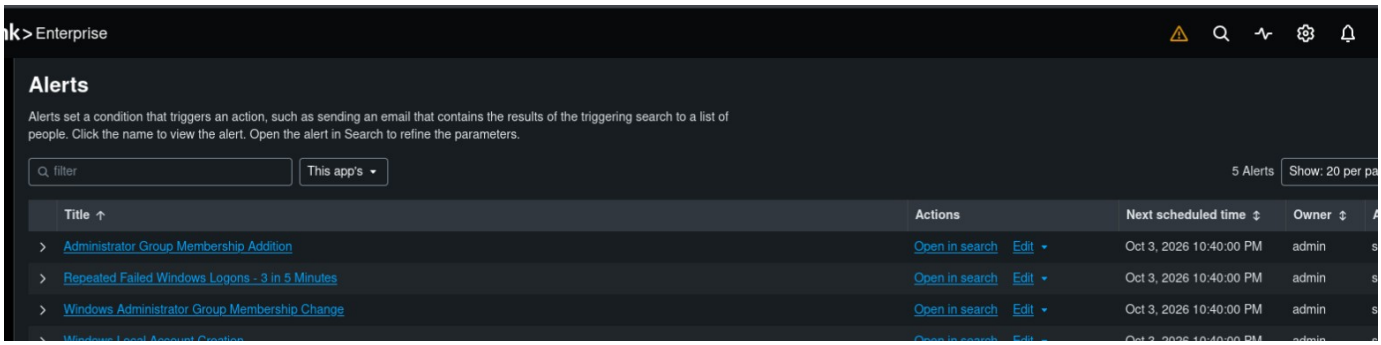
Key Investigation Outcome

Detection	3 failed logons in 5 minutes
Account / Host	labuser / DESKTOP-4AMTH10
Source / Logon Type	127.0.0.1 / Interactive (Type 2)
Correlation	3 failed 4625 events followed by successful 4624 logon
Process Review	Event ID 4688 reviewed; no labuser PowerShell execution found in the investigated 30-minute window
Disposition	True Positive - Authorized/Benign Security Test

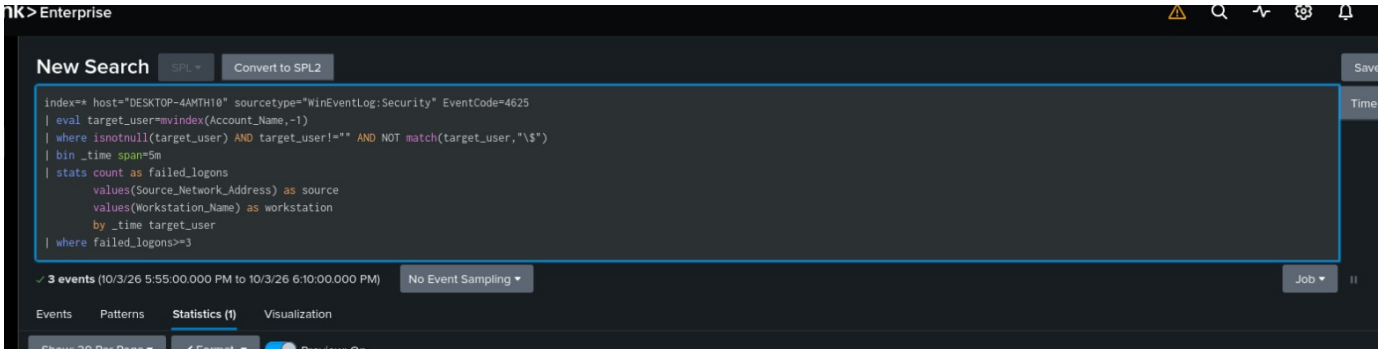
SOC Monitoring & Investigation Evidence



1. Monitoring - SOC Windows Security Monitoring Dashboard at final validation



2. Detection - Five enabled Splunk security alerts at project completion



3. Investigation - Triggered failed-logon detection: labuser, 3 failures, source 127.0.0.1, workstation DESKTOP-4AMTH10

Skills Demonstrated

Splunk Enterprise • Splunk Universal Forwarder • SPL • Windows Security Event Logs • Detection Engineering • Scheduled Alerts • Alert Suppression • SOC Dashboards • Log Analysis • Alert Triage • Incident Investigation • Authentication Correlation • Event ID 4624 • Event ID 4625 • Event ID 4688 • Kali Linux • Ubuntu Server • VMware Workstation

Project Scope

Personal controlled cybersecurity home lab completed in October 2026. This project demonstrates hands-on SOC workflow practice at lab scale and is not presented as production SOC employment.